

In [3]: `import numpy as np`

In [4]: `def simpson_two_slices(f, x_i, h):
 """
 Simpson's rule over [x_i-h, x_i+h]: (h/3)(f_{i-1} + 4f_i + f_{i+1}).

 f: f(x)
 x_i: center point of the 2 slices
 h: space between nodes
 """
 return (h/3) * (f(x_i - h) + 4*f(x_i) + f(x_i + h))`

In [6]: `# verify the derivation:
1. fit a quadratic through the 3 points
2. check it matches the Simpson formula

test function, evaluation point and step size
f = np.sin
x_i = 1.0
h = 0.3

3 nodes: x_i-h, x_i, x_i+h
xs = np.array([x_i - h, x_i, x_i + h])

solve for a, b, c of the quadratic a*x^2+b*x+c
a, b, c = np.polyfit(xs, f(xs), 2)

exact integral of quadratic from x_i-h to x_i+h
integral = (a/3)*((x_i+h)**3 - (x_i-h)**3) + (b/2)*((x_i+h)**2 - (x_i-h)**2) + c*(2*h)

print results
print("quadratic integral:", integral)
print("Simpson: ", simpson_two_slices(f, x_i, h))`

```
quadratic integral: 0.49736598118864683
Simpson:    0.49736598118864694
```